

# SHUBHAM KISHOR KALE

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*AI Engineer building production LLM systems, agentic RAG with citation checks and evals, and the data platforms they run on.*

## WORK EXPERIENCE

Quantegy Analytics | New York, USA

May 2025 - Present

*Analytics / AI Engineer*

- Shipped an **LLM-powered navigation assistant** into a production analytics platform, now answering 150+ questions a day; routed 46 of 50 randomly sampled questions to the right dashboard page.
- Automated intake of 100+ client requests a month with an idempotent **Dagster** pipeline: an LLM extracts each email into Pydantic-validated JSON; 29 of 30 randomly sampled tickets passed human review unedited.
- Ranked 420 stores for endcap placement with a **stacked tree and linear-regression ensemble**; the 21 top-ranked stores given endcaps posted +\$118K in week-1 sales on a \$984K base (+12%) vs +0.25% in 14 controls.
- Built the **Snowflake data platforms** for 3 clients on Dagster and Airflow: 4M+ healthcare records from Redshift, 120+ EHR tables for a correctional healthcare client, ~5x faster ingestion for an insurance client.

Mettler-Toledo | Mumbai, India

Mar 2024 - Jun 2024

*AI Engineer Intern*

- Designed a **RAG system with fine-tuned T5/BART models** that drafts KPI commentary from 10M+ financial records; 85%+ of drafts rated relevant in a 40-case team review, saving 15 hrs/month of manual write-ups.

Marcellus Investment Managers | Mumbai, India

Mar 2023 - Nov 2023

*Quantitative Analyst Intern*

- Developed a **regression-based impact-cost model** (R-squared 0.82) for trade-execution optimization and a **forensic-screening ML ensemble** (84% accuracy, 87% specificity) adopted into fraud-risk screening.

## PROJECTS

**AI Financial Research Agent** · [GitHub] · [Live Demo] · *Agentic RAG, MCP, Evals, Observability*

- Architected a **LangGraph ReAct agent** over SEC 10-K filings with 5 MCP-served tools, including XBRL fact lookup, and per-thread conversation memory (11 of 11 follow-ups in a small probe, 3 of 11 without).
- Validated each served answer with a **fail-closed contract**, no model in the loop: figure-to-source matching and citation validation against the retrieved evidence, passed by all 71 benchmark answers.
- Lifted retrieval hit@5 from 0.51 to 0.66 on the same benchmark with cross-encoder reranking; XBRL fact tools raised figure accuracy from 73% to 100% on its 45 figure items and **RAGAS faithfulness from 0.83 to 0.94**.
- Engineered the full stack: FastAPI streaming cited answers over Server-Sent Events to a Next.js UI, **LangSmith tracing** and a per-query meter (p50 2.8 s, p95 6.9 s, about \$0.002); 362 tests, including 33 covering failure paths.

**AI Auditor** · [GitHub] · [Live Demo] · *Human-in-the-Loop, Multimodal Document Extraction, Guardrails*

- Deployed a 5-stage audit review workspace (FastAPI, React, Supabase) where a **vision LLM** proposes facts and judgments and deterministic Python owns materiality, sampling and approval, so the model never signs off.
- Implemented **structured extraction** over PDFs, scans and spreadsheets through a strict JSON-schema contract with a vision model, screened by 16 deterministic, policy-cited evidence checks before a fact reaches a workpaper.
- Enforced controls in code: **row-level security** on 7 Postgres tables, row-locked versioned approvals, a staleness cascade on corrections, and a model adapter with retries and backoff; 62 tests, 9 of them security regressions.

## TECHNICAL SKILLS

**Languages, Frameworks & Cloud:** Python, SQL, TypeScript, PyTorch, scikit-learn, Hugging Face, Claude API, Gemini API, LangChain, LangGraph, LangSmith, FastMCP, ChromaDB, RAGAS, FastAPI, Next.js, Docker, AWS (S3, Redshift), Snowflake, PostgreSQL, Pydantic, pytest, GitHub Actions (CI/CD), Claude Code, Codex

**Concepts & Methods:** Generative AI (RAG, LLM Agents, MCP, Fine-tuning), Structured Outputs, LLM Evaluation, LLM Observability, LLM Guardrails, Prompt Engineering, Machine Learning

## EDUCATION

Stony Brook University, New York, USA | *M.S. in Data Science* | **GPA: 3.78/4**

Aug 2024 - Dec 2025

NMIMS University, Mumbai, India | *B.Tech. in Data Science*

Jul 2019 - May 2023